LS-CAT Management Board Meeting

December 21, 2018 10:30 AM CT (11:30 AM ET)

We'll open the conference up a half hour early to test connections, solve problems, etc. Please feel free to join us.

To join the meeting on a computer or mobile phone visit https://bluejeans.com/8474912985

Phone Dial-in

1.408.740.7256 (US (San Jose))

1.888.240.2560 (US Toll Free)

1.408.317.9253 (US (Primary, San Jose))

Global Numbers: http://bluejeans.com/numbers

Meeting ID: 847 491 2985

Room System
199.48.152.152 or bjn.vc

Meeting ID: 847 491 2985

Please connect from a quiet space using a phone handset, headphones, or earbuds. Speakerphones and computer speakers cause echos.

Keep your microphone muted unless you are talking.

Agenda

10:00 Sound check / connection debugging

10:30 Welcome: minutes and agenda approval (Stuckey)

10:35 Brief Status / Publications and Usage Reports (Brister)

10:45 Brief APS / APS-U update (Streiffer)

10:55 LS-CAT Downtime planning (Brister)

11:15 Future Directions Committee Report (Forest)

11:30 New Business

Brief LS-CAT Status

APS experienced had some downtime due to vacuum issues

ANL implemented changes in how email from anl.gov works killing our processes for servicing users mid run (since fixed)

Indexing and strategy available to remote users (via data tab on remote interface)

Raster scanning implemented on F and G (step scans)

Raster scanning mostly implemented on D (horizontal fly scan, vertical step scan)

Bob Fischetti (GM/CA Group Leader / APS Life Sciences Advisor) is arranging for Dectris to offer substantial discounts for extended warranties. The undiscounted price is $75K / year for our Eiger 9M. I have not yet seen a discounted quote but I suggest that we participate as these warranties are expensive. Additionally, setting a precedence for this sort of collaboration is a good idea.

Publications/Dissertations/Usage

Lists of publications and dissertations are now being generated for each member along with usage information for each run. Publications are current through 2017 and usage data is current through the 2018-2 run. Each of you has been sent a link for your institution's spreadsheets.

Starting next month we'll start adding pdb entries as well as updating the lists to cover 2018.

If your institution has requested usage information this is the spreadsheet we'll provide.

Publication lists are taken from the APS database as well as the PDB.

Thank you everyone who has provided us with updated dissertation lists, these remain the most difficult part of this for us to track down. For those who have yet to explore this, please do.

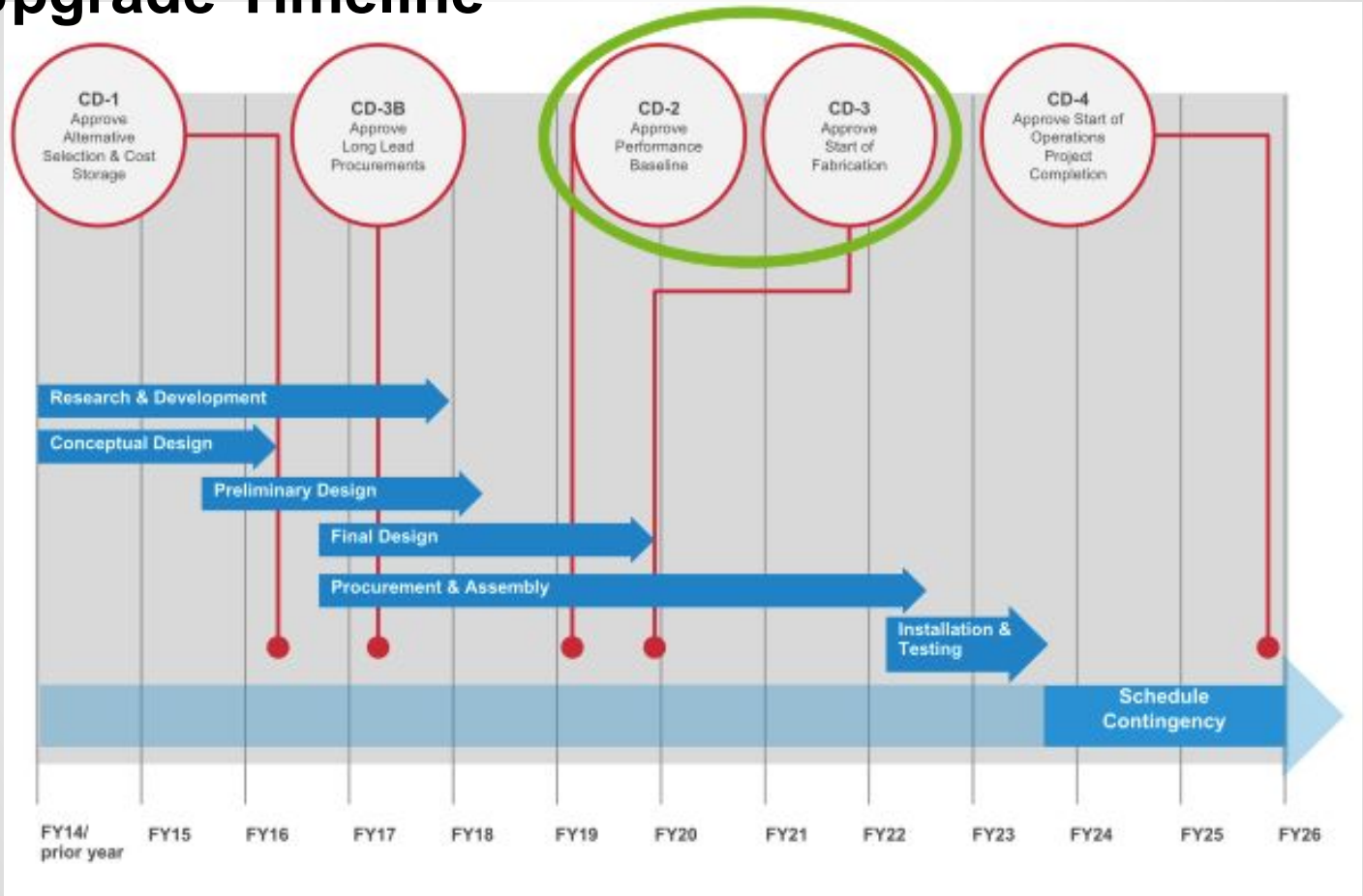
LS-CAT Upgrade Planning

APS Upgrade Timeline

12-ish month dark period starts in 2022

Operations should start up in 2023

Full operations 2024-ish



		Priority	Funded	Worst Case	Item	Est Cost	Contingency
Total of all Costs	$8,928,000	1	1	1	Move Inboard Beam masks downstream and lengthen bellows	10,000	2,500
Maximum Priority	9	2	1	1	Remove side-bounce mono's water to vacuum braze (2 each)	30,000	7,500
Priority Costs	$438,000	3	1	1	Remove commissioning window: install differential pump	10,000	2,500
		4	1	1	Service Cryocooler (1 each)	5,000	1,250
Worst Case Cost Contingency	25%	5	1	1	Service Cryo-Tigers (10 each)	75,000	18,750
		6	1	1	Service Cryo-Jets	5,000	1,250
Funding Level	$500,000	7	1	1	Upgrade Motor Controllers	50,000	12,500
Max priority w/contingency		8	1	1	Service D-Mono Electro-Mechanics	3,000	750
		9	1	0	Upgrade D MD-2	250,000	62,500
		10	0	0	Upgrade side-bounce monos	200,000	50,000
		11	0	0	New D-Mono Cooling	20,000	5,000
		12	0	0	New D-Mono Crystals	20,000	5,000
		13	0	0	Upgrade F MD-2	250,000	62,500
		14	0	0	Upgrade G MD-2	250,000	62,500
		15	0	0	New Cryocooler	300,000	75,000

Upgrade Planning

What follows is the outline of a the next 5 years of LS-CAT starting at 2024 and working backwards.

Optimistic predictions here.

Pessimistic predictions here.

The Vision of 24

- $3M per station of upgrades ($2M for detectors and $1M for diffractometers, robotics, and optics upgrades)
- Possible nanoprobe with sample preparation facilities providing the ultimate resource for LS-CAT members use of all the nanoprobes at the APS

- Minimal changes to Sector 21 to make it plausibly ready for upgraded APS
- Minimal servicing of detectors.
- No new hardware capabilities
- Much hardware obsolete
- Start rebuilding staff

The Excitement of 23

Beginning of commissioning:

- Not much is working yet.
- The beam is less than perfect but there is promise.

Beginning of commissioning:

- Not much is working yet.
- The beam is less than perfect but better than we saw it last.



The Panics of 22

1. APS goes down with only 6 months notice of the exact date.
2. Staff crystallographers support members at other facilities and help with analysis of data already collected.
3. Shinny new LS-CAT Agreement allows for new partners such as SBC
4. Preparations made to accommodate new detectors
5. Diffractometers sent out for upgrades
6. Serial crystallography support finalized

1. APS goes down with only 6 months notice of the exact date.
2. Without full funding, half of the remaining LS-CAT staff leaves.
3. LS-CAT Agreement expires. Presumably any members remaining would band together to maintain a presence at Sector 21.



The Crises of 21

1. A Fiscal 2022 budget is approved. Funding for an upgraded LS-CAT is approved
2. A new LS-CAT Agreement is signed by all current members
3. The current LS-CAT manager hands over the reins to a fully qualified new LS-CAT manager.

1. A minimal Fiscal 2022 budget is approved. This covers the start of the dark period and only a minimal staff.
2. Amid a great effort to renew or replace the LS-CAT Agreement it is discovered that some of the members have decided not to participate leaving the previously approved budget partially unfunded.
3. The current LS-CAT manager leaves. Interim manager named.

The Labors of 19-20

1. A new LS-CAT Agreement is hashed out and starts the rounds
2. Planning for both a fully funded upgrade and an unfunded non-upgrade get fleshed out
3. A new LS-CAT Management Plan is written and accepted by the APS. This is an agreement between the APS and LS-CAT that allows us to execute the new LS-CAT Agreement and meet the requirements of the MOU
4. In the case of SBC joining LS-CAT, an interface document is developed and approved by LS-CAT, SBC, and APS. This document lays down the interaction and responsibilities between the various parties
5. A decision on how to proceed (or not) with 21-ID-E is made.
6. The disposition of the upstream end of 21-ID-D is decided.

Charge to Committee

1. Develop the “Vision of 24”
2. Develop a plan to fund that vision
3. Suggest changes to the LS-CAT Agreement
4. Give feedback on the aforementioned Interface Document prior to it being presented to the Management Board

Assets

Asset	Description	Upgrade Risks
Staff	7.5 FTEs	Some attrition likely
Annual Budget	$1.7M	Difficult to maintain during dark period. VPRs do not want increased dues when a member drops out. Unlikely all 8 current members will remain.
Capital Fund	~$1M	Not enough to maintain staffing in lieu of significant member contributions
Intellectual Infrastructure	Culture Procedures Software	Maintained through staff continuity and thus is threatened by attrition and/or RIF.
Physical Infrastructure	Sector 21 and environs	Aging but still serviceable. Detectors need replacement and diffractometers need upgrades. Robotics and optics need attention as well.

General User Time

Bad News: After the APS upgrade CD-4 (~2026) LS-CAT will be required to "share" 50% of our beamtime with general users. The current GU fraction is 25%.

Good News: CAT time can be treated as GU time if certain conditions are met:
1) There is a GU proposal in place
2) The proposal is rated highly enough that the Beamtime Allocation Committee assigns time.

Note that the required 50% fraction is after an 8% allowance for configuration and alignment which more that we currently need. Hence, the GU portion of 5000 total running hours will be 2300 hours, not 2500 hours.

The Plus Side: Members would not be required to worry about dealing with the GU system as the LS-CAT staff can generate a proposal for each member. We would keep improving the proposals and resubmit them until we have a critical mass of 9% of our total CAT usage recast as GU usage

The Plus Plus Side: LS-CAT members would qualify for time in excess of their LS-CAT allocation and the proposals could be used to receive time at other beamlines.

The Minus Side:
1) A lot of busy work to achieve what we have already.
2) If not enough LS-CAT member time can be cast as GU time then up to 9% of member time is at risk.
3) We'll need to get used to using more of the night shifts for members. The current 10AM-10PM staff coverage will have to be stretched on each end.

SBC as an LS-CAT Member

LS-CAT scheduling is based on dividing the CAT share of the beamtime among 6.5 shares. Currently we service only 5 shares leaving 1.5 available for SBC. If the APS were to allow SBC to control the nominal 25% GUP time then they would have 41% LS-CAT time which is enough for them to make a go of things.

The capital buy-in for LS-CAT members was $2M in 2002 dollars which translates to about $2.8M today so the nominal capital buy-in for SBC would be $4.2M. The annual dues would be equivalent to 1.5 shares or $552K at today's rates. This is the equivalent of 2 full time scientists plus some using Northwestern's benefits and off-campus overhead rates.

It is unlikely that ANL will cut a check to NU for any of these amounts so payment in kind will need to be worked out.

Requirements for an LS-CAT/SBC deal:

1) Sources of funding identified
2) A new LS-CAT management plan that codifies the SBC contributions to LS-CAT
3) A comprehensive plan specifying how SBC and LS-CAT interact listing who is responsible for what
4) Approval of the LS-CAT Management Board
5) A new (or modified) LS-CAT Agreement that takes into account this new membership

How to deal with the 50% GU requirement after the APS upgrade project finishes

Current requirement is 25%

Members will be allowed to submit general user proposals and compete for that 50% piece of the pie

LS-CAT staff can ghost write proposals

These GU proposals can be used to obtain time at other beamlines to simplify access to all APS facilities

	Hours/Year/Station	Days/Year/Station	Days/Run/Station
Gross Machine Availability	5,000	208.3	69.4
8% Alignment Set Aside	400	16.7	5.6
Net Machine Availability	4,600	191.7	63.9
Current GUP Allocation	1,150	47.9	16.0
Current CAT Allocation	3,850	160.4	53.5
Per Share (6.5 Shares Total)	592	24.7	8.2
Current LS Need (5 Shares)	2,962	123.4	41.1
Time Remaining	2,038	84.9	28.3
Percent of Total Remaining	41%		
Percent of Net Remaining	44%	⇐ 6% away from GU goal!	
GU Required (50%)	2,300	95.8	31.9
CAT/GU Needed	262	10.9	3.6
Share CAT/GU Needed	52	2.2	0.7

New LS-CAT Agreement Suggestions

1. Allow institutions to sign individually. NU sends separate signing sheets to each institution in parallel instead of one sheet serially
2. Specify annual budget (with inflation/deflation annual corrections) with a mechanism for one-time costs and/or changes in excess of contracted amount
3. Embrace fractional shares
4. Codify third party interactions to account for APS requirements
5. Have management board votes in proportion to financial contributions
6. Have explicit APS/ANL representation on management board